

GYAN SETU

Chapter-2, Coordinate Geometry

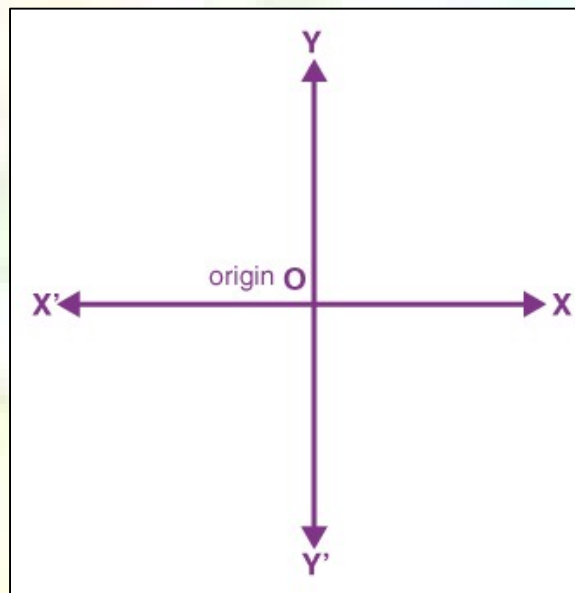
1. Coordinate Geometry

Coordinate Geometry is an important branch of mathematics that connects Algebra and Geometry. It allows us to locate points on a plane using ordered pairs and has applications in higher mathematics like Trigonometry and Calculus.

2. Cartesian System

The Cartesian system is a method of describing the position of a point in a plane using two perpendicular lines:

- The horizontal line is called the X-axis.
- The vertical line is called the Y-axis.
- These lines intersect at a point called the Origin (O).



3. Origin

The Origin (O) is the point where the X-axis and Y-axis intersect. It is the reference point for measuring distances:

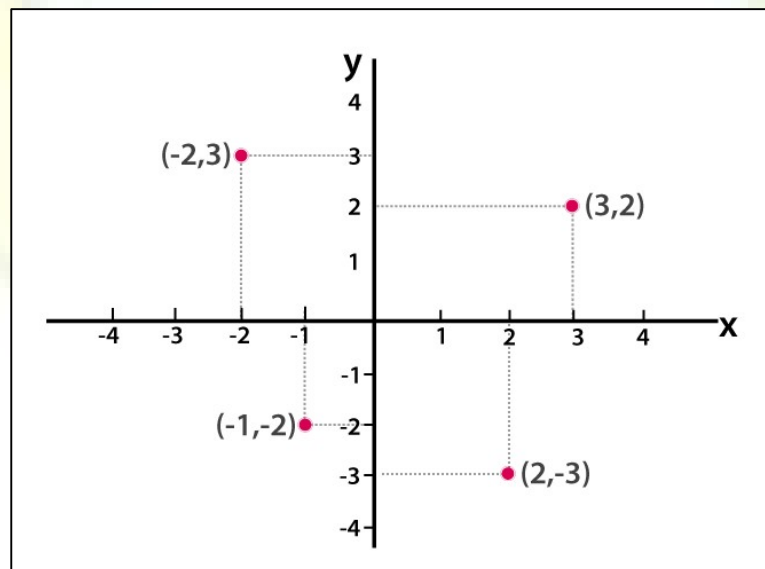
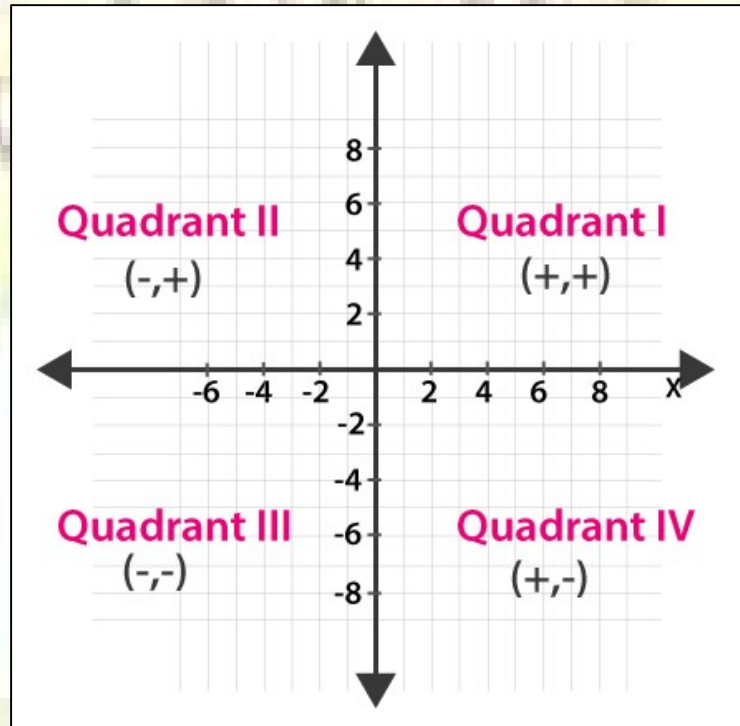
- Positive directions: OX and OY.
 - Negative directions: OX' and OY'.
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4. Coordinate Axes and Quadrants

The plane is divided into four parts, called Quadrants, by the X and Y axes. The quadrants are numbered I, II, III, and IV in the anti-clockwise direction.

Signs of coordinates in quadrants:

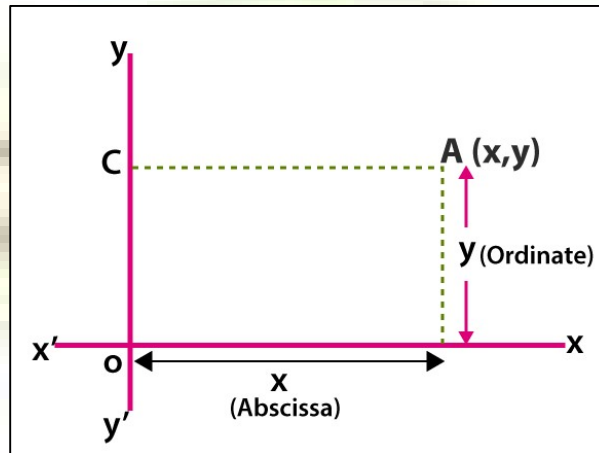
- I Quadrant: $(+x, +y)$ e.g., $(3, 2)$
- II Quadrant: $(-x, +y)$ e.g., $(-2, 3)$
- III Quadrant: $(-x, -y)$ e.g., $(-1, -2)$
- IV Quadrant: $(+x, -y)$ e.g., $(2, -3)$



5. Coordinates of a Point

A point in the plane is represented as an ordered pair (x, y) :

- x is called the Abscissa (distance from Y-axis).
- y is called the Ordinate (distance from X-axis).



Example: Point $(3, 2)$ is 3 units away from the y-axis and 2 units away from the x-axis.

6. Important Notes

- $(x, y) \neq (y, x)$, i.e., the order of coordinates matters.
 - Each point corresponds to one unique ordered pair.
 - A point lying on the X-axis has $y = 0$, e.g., $(-1, 0)$.
 - A point lying on the Y-axis has $x = 0$, e.g., $(0, 3)$.
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